

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular End Semester Examination – Summer 2022-23

Course: B. Tech.

Branch : Civil

Semester : IV

Subject Code & Name: BTCVC402/BTCIC402 (Environmental engineering)

Max Marks: 60

Date: 15/07/2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

- A) The population data for a town is given below Estimate population at the end of the year 2020 by arithmetical increase method. CO1 6

Year	1950	1960	1970	1980	1990
Population	35000	37500	43500	52000	57500

- B) Explain the characteristics of the water in short. CO1 6
- C) What is water demand and explain various water demand CO1 6

Q.2 Solve Any Two of the following.

- A) Explain the purpose of aeration of water with a neat sketch of cascade type of aerator. CO2 6
- B) What is the difference between slow sand filter and rapid sand filter CO2 6
- C) Design a rectangular settling tank to treat 2 MLD of water . CO2 6

Assume detention time of 3 Hrs and flow through velocity of 7.5cm/min. if the depth of tank is 3m, find the overflow rate and dimensions of the tank.

Q. 3 Solve Any Two of the following.

- A) Compare between Continuous water supply system and Intermittent water supply system. CO3 6
- B) Explain the following layout system for water distribution CO3 6
- (a) Dead end system.
- (b) Radial system.
- C) Which factors consider in the design of Distribution system CO3 6

Q.4 Solve Any Two of the following.

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|---|------------|----------|
| A) Explain physical, chemical and biological characteristics of municipal waste water. | CO4 | 6 |
| B) Write a note on sewerage system and explain any one of them in detail. | CO4 | 6 |
| C) Explain E-waste disposal method in short. | CO4 | 6 |

Q. 5 Solve Any Two of the following.

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| A) What are the effects of lapse rate on plume behaviour. | Remember | 6 |
| B) State the method of control of particulate matter. | Remember | 6 |
| C) What are the major sources of air pollution. | Remember | 6 |

***** End *****